

Liminal

Zhitao Lin
Peabody Institution
Johns Hopkins University
USA
zlin38@jhu.edu

Na Xing
Johns Hopkins University
USA
xxninaxn@hotmail.com

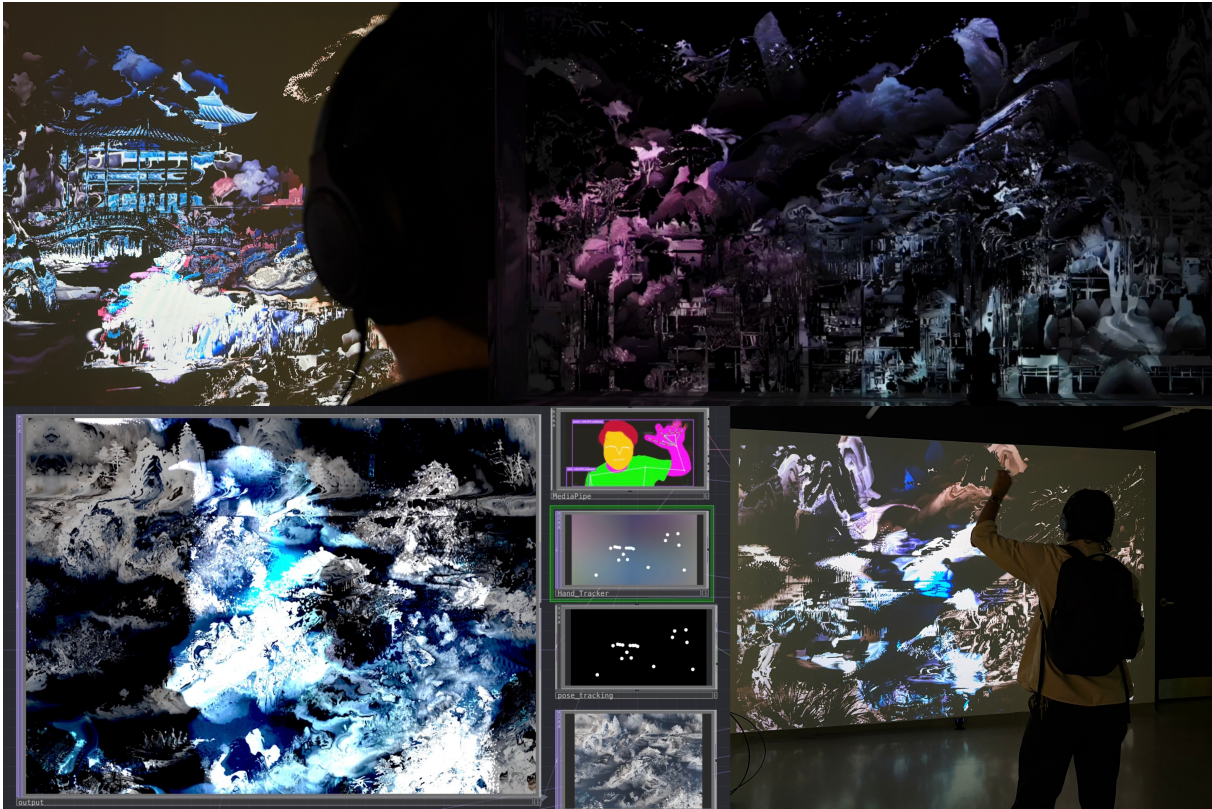


Figure 1: Liminal

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1 Overview

Liminal is an AI-assisted interactive audiovisual installation that merges gesture, sound, and image into a unified realtime experience. The work invites participants to explore how music and visuals can emerge organically from human movement.

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Using computer vision (CV), the system captures the participant's body motions through a camera and translates them into coordinate data. These data activate AI-generated digital ink landscapes and ludian-inspired visuals, enhanced by dynamic real-time effects such as glitch textures and particle flows.

At the same time, the motion information drives a real-time sound synthesis engine that transforms gesture into musical parameters—pitch, duration, and spatial position. All sonic materials are generated live, framed within the aesthetics of traditional Chinese music, allowing ancient timbres and modern synthesis to blend seamlessly. In Liminal, every element—painting, visuals, and music—is created in the moment, continually reshaping itself in response to human gesture. Rather than a fixed composition, it is a generative environment—a dialogue between technology and intuition, where creation begins the instant the participant moves.